
// BIOKEY SYSTEM v2.0

Your heartbeat is the key — at any stress level.

Stress-Invariant Biometric Authentication · AES-256-GCM

HackTM 2026 // Defence Track

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Authentication breaks when it matters most.

PASSWORDS

Forgotten under stress.
Coerced under threat.

SMART CARDS

Lost in the field.
Stolen with the operator.

FINGERPRINTS

Fail with gloves.
Fail when injured.

BIOMETRICS AT REST

ECG at 160 bpm $\neq$ ECG at 70 bpm.
System denies its own operator.

→ In the critical moment, the system locks out the right person.

*"ECG is unique and always present.
The problem is that nobody enrolled
the operator under stress."*

THE PROBLEM

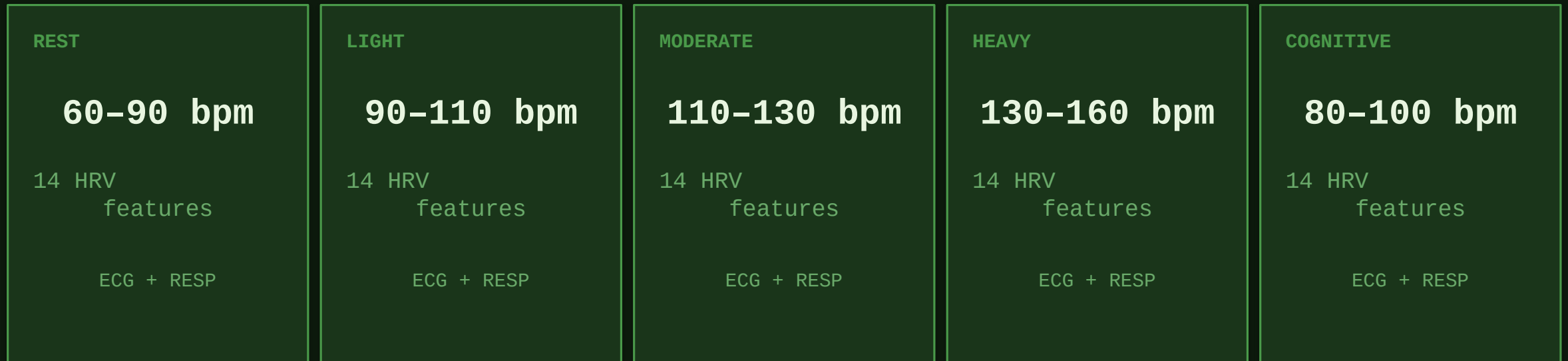
Soldier at rest: HR ~70 bpm
Under combat stress: 140-160 bpm
ECG morphology shifts → single template
fails

THE SOLUTION

Enroll the operator at ALL stress levels.
System recognizes at 60 bpm and 160 bpm.
No manual state input required.

Multi-Template Stress Enrollment

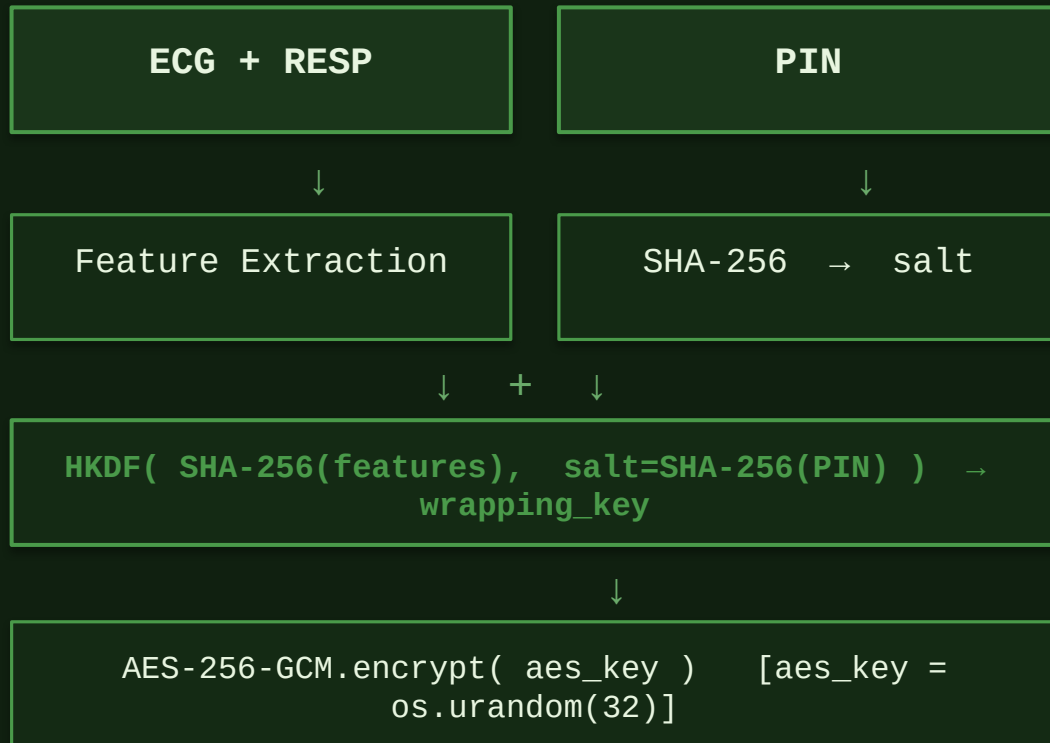
Like a treadmill stress test at the cardiologist – capture biometric signature across the full physiological spectrum.



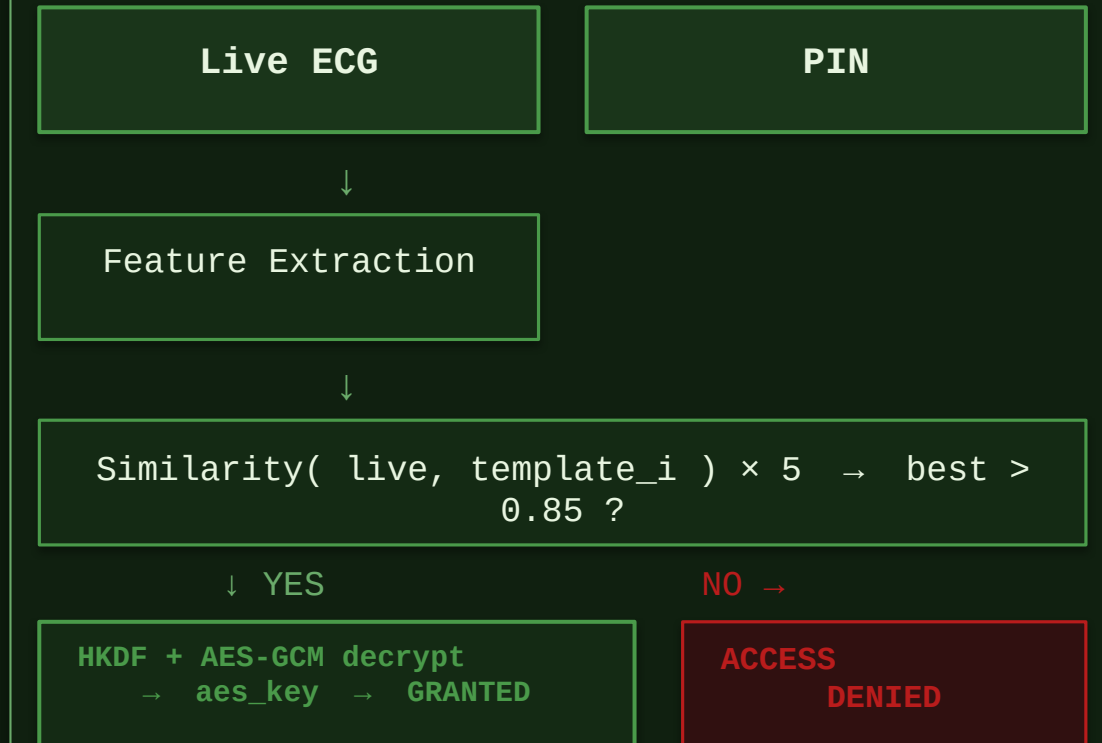
→ At authentication: live ECG + PIN → nearest template matched automatically → AES-256 key unwrapped

From ECG to AES-256 Key

ENROLLMENT



AUTHENTICATION



What the attacker needs – and why they fail.

CAPTURED HARDWARE

Terrain DB is AES-256-GCM ciphertext.
Without ECG – permanently inaccessible.

WRONG OPERATOR

Similarity $< 0.85 \rightarrow$ wrong wrapping key
 $\rightarrow$ AES-GCM tag mismatch $\rightarrow$ denied.

CORRECT ECG, WRONG PIN

Similarity passes.
PIN salt missing $\rightarrow$ cryptographic failure.

STRESS STATE MISMATCH

5 templates cover the full spectrum.
Nearest match selected automatically.

KEY NEVER STORED

aes_key exists only in RAM.
Derived fresh each authentication.

BRUTE FORCE

AES-256: 2^{256} keyspace.
Computationally infeasible.

// PRODUCTION NOTE

In production: biometric profile stored in TPM/HSM on-device. Key derivation inside secure enclave – profile never exposed in memory. Current implementation demonstrates the cryptographic layer (HKDF + AES-256-GCM), identical regardless of storage backend.

System Extensions.

LAYER 1

Biometric – ECG + Respiration

Stress-invariant multi-template enrollment. Unique cardiac signature across all physiological states.

LAYER 2

PIN – Second Factor

HKDF salt derived from SHA-256(PIN). Correct biometrics + wrong PIN = cryptographic failure.

LAYER 3

Random Seed – Independent AES Key

`aes_key = os.urandom(32)` – not derived from biometrics. Full biometric compromise cannot reconstruct the key.

LAYER 4

Two-Man Rule

Two operators must authenticate simultaneously. Neither can unlock the system alone. NATO nuclear standard.

LAYER 5

Dead Man's Switch

Key expires if operator does not re-authenticate every N minutes. Capture or incapacitation = automatic revocation.

Grounded in research – filling the gap.

LITERATURE CONFIRMS OUR FOUNDATION

ECG is person-distinctive

Biel et al. – "ECG Analysis: A New Approach in Human Identification" – IEEE Trans. 2001

→ validates that ECG uniquely identifies individuals

ECG → crypto key is feasible

Venkatasubramanian et al. – "EKG-based Key Agreement in Body Sensor Networks" – IEEE 2008

→ confirms 49.99% Hamming distance between persons

Physiological signals for security

Poon et al. – "A Novel Biometrics Method to Secure Wireless Body Area Sensor Networks" – 2006

→ proves IPI-based keys are random, time-variant, person-distinctive

THE GAP WE FILL

All existing schemes:

Single enrollment at rest.
Fail under physiological stress.

None address:

Operator authentication at
variable stress levels.

BioKey contribution:

5-template stress-invariant enrollment.
HKDF + PIN second factor.
AES-256-GCM – NATO standard.

Any role where the right person must be at the controls – right now.

MILITARY / DEFENCE

- Drone operator authentication
- Pilot / co-pilot verification
- Nuclear launch authority (two-man rule)
- Forward operating base access control

CIVIL / INDUSTRIAL

- Surgeon – operating room access control
- Air traffic controller verification
- Nuclear plant operator authentication
- Critical infrastructure access control

Same platform. Same cryptographic guarantees. Different operator profiles.

*"Your heartbeat is the key –
at any stress level."*

AES-256-GCM · HKDF-SHA256 · Multi-Template HRV · PIN Second Factor

github.com/IosifSramec/biokey

HackTM 2026
